

MacDonald-Thorne Black-hole Electrodynamics for Nerves. Part I: Motivation

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Motivation

- ▶ We have a system of partial differential equations that govern axon-axon interaction and coupling in three dimensional Euclidean space and time [1].
- ▶ We wish to see what happens when we extend this system to a curved spacetime setting - on rederivation, will it then interact with gravitational waves and universal expansion, however weakly?
- ▶ This has been done in [2] and [3]. However, here we are motivated by a possibility of a more rigorous reformulation.
- ▶ To this end, we have picked [4] since it gives a good introduction to the relation between 3-space + 1-time and 4-spacetime formulations

1. Introduction

- ▶ Pulsars were discovered in 1968 [5]
- ▶ These are rotating neutron stars
- ▶ The rotational energy is transmitted to radiating particles by a strong magnetic field embedded in the star
- ▶ The corresponding theoretical formulation is called 'pulsar electrodynamics' [6]

Quasars

- ▶ Quasars were discovered four years before pulsars [7]
- ▶ Black holes can energize quasars [8] [9] [10]
- ▶ The mechanism is essentially the same as in the pulsar case
- ▶ Magnetic fields transmit rotational and orbital energy to distant, radiating particles

Discomfort with Curved Spacetime

- ▶ The theory of quasars took 13 years to develop
- ▶ The authors of [4] speculate why this long time would have been needed
- ▶ The reason is that curved 4-D spacetime is needed as an abode for crucial parts of the theory
- ▶ Many astrophysicists are uncomfortable with curved spacetime
- ▶ Relativists on the other hand, are quite comfortable
- ▶ Relativists prefer to think about physics in a geometric, frame-independent way, representing the electromagnetic field by a tensor F (a tensor is a higher-dimensional matrix), a four-dimensional geometric object.

Discomfort (contd.)

- ▶ The astrophysicist, on the other hand, prefers to split this tensor into a 3D electric field E and magnetic field B
- ▶ This sacrifices the 'general covariance' of the theory - a property we may take up later
- ▶ But it allows a comparison with the flat-spacetime theory of pulsar electrodynamics

Discomfort (contd. 2)

- ▶ The relativist likes to regard spacetime as a global, four-dimensional manifold, free of any global reference frames
- ▶ The astrophysicist's intuition is based on a fixed, absolute 3-D space with an associated universal reference frame, and a universal time.
- ▶ The relativist's viewpoint is essential when spacetime is highly dynamical

MacDonald-Thorne Formulation in [4]

- ▶ In stationary, but curved spacetimes, one can reformulate electrodynamics in terms of an absolute but curved three-dimensional space and a universal time
- ▶ The familiar variables of electrodynamics appear in this formulation: $\mathbf{E} \rightarrow$ electric field, $\mathbf{B} \rightarrow$ magnetic field, $\rho_e \rightarrow$ charge density and $\mathbf{j} \rightarrow$ current density
- ▶ In [4] Section 2, the authors present without derivation, the absolute-space/universal-time formalism, restricted to the space outside a rotating, axially symmetric black hole.

Next Steps

- ▶ Section 2.1: mathematical structure of the formalism, and its physical interpretation
- ▶ Section 2.2: the equations of electrodynamics, local laws of energy balance and force balance, global laws of conservation of angular momentum and 'redshifted energy'
- ▶ Section 2.3: the Znajek-Damour 1978 theory [11] of boundary conditions of the black hole's horizon, reformulated in the absolute space language

References

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